

# Truong Buu Phan

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[Google Scholar](#) - [LinkedIn](#)

## RESEARCH INTERESTS

I work at the intersection of information theory, tokenization, and machine learning. My research develops information-theoretic and statistical methods for neural compression and tokenization, targeting efficient video/image compression, language modeling, and scalable AI systems. Overall, I enjoy developing new foundational tools for the next generation of efficient, reliable, and trustworthy AI systems.

*Research Topics:* LLMs, tokenization, information theory.

## ACADEMIC BACKGROUND

*Ph.D. Electrical and Computer Engineering* 2022 - Oct 2026  
University of Toronto, Canada

- Advisor: Prof. [Ashish Khisti](#).
- Research topic: sampling, compression and representation learning.
- Thesis: Probabilistic and Coding Techniques for One-Shot Compression and Language Modelling.

*MASc. Electrical and Computer Engineering* 2017-2019  
University of Waterloo, Canada

- Advisor: Prof. [Krzysztof Czarnecki](#).
- Research in Bayesian deep learning and out-of-distribution detection.

*BEng. Electrical Engineering* 2012-2016  
Vietnam National University, Vietnam.

## RESEARCH EXPERIENCE

*Microsoft Research, USA* June 2026 - Aug 2026  
Research Intern.  
Host: [Dr. David Alvarez-Melis](#).

*Qualcomm AI Research, Canada* May 2025 - Aug 2025  
AI Research Intern.  
Host: [Dr. Roland Memisevic](#).  
Project: Length Generalization in Hybrid Recurrent-Transformer.

*Meta AI (FAIR), USA* Mar 2024 - Oct 2024  
AI Research Intern (Language Model and Probabilistic Reasoning).  
Host: [Dr. Karen Ullrich](#).  
Project: Probabilistic reasoning, tokenization, LLMs.

*LG Electronics AI Lab, Canada* May 2021 - May 2022  
Research Engineer  
Project: Automated check-out and neural symbolic AI.

*[Algolux](#) (acquired by Torc Robotics in 2023), Canada* Oct 2019 - May 2021  
Research Scientist  
Collaborators: Prof. [Felix Heide](#) and Dr. Fahim Mannan.  
Projects: Adversarial attack and robust vision for autonomous driving.

**SELECTED  
PUBLICATIONS**

For full publications, see my [Google Scholar](#).

**Buu Phan**, Gergely Flamich, Ashish Khisti and Shahab Asodeh. “Multi-Marginal Couplings for Metropolis-Hastings ” (**Under Review**)  
<https://arxiv.org/abs/2605.12807>.

**Buu Phan**, Ashish Khisti and Karen Ullrich. “Cross-Tokenizer Likelihood Scoring Algorithms for Language Model Distillation” (**International Conference on Learning Representations 2026**)  
<https://arxiv.org/pdf/2512.14954>.

Joseph Rowan, **Buu Phan**, Ashish Khisti. “List-Level Distribution Coupling with Applications to Speculative Decoding and Lossy Compression” (**Conference on Neural Information Processing Systems 2025**).  
<https://arxiv.org/abs/2506.05632>.

**Buu Phan**, Reza Ebrahimi, Sanjay Haresh, Roland Memisevic. “Delayed Attention Training Improves Length Generalization in Transformer–RNN Hybrids” (**What Can(t) Transformers Do? Workshop @ Conference on Neural Information Processing Systems 2025**).  
<https://arxiv.org/abs/2510.00258>

**Buu Phan**, Brandon Amos, Itai Gat, Marton Havasi, Matthew Muckley and Karen Ullrich. “Exact Byte-Level Probabilities from Tokenized Language Models for FIM-Tasks and Model Ensembles” (**International Conference on Learning Representations 2025**). <https://arxiv.org/abs/2410.09303>.

**Buu Phan**, Ashish Khisti. “Channel Simulation and Distributed Compression with Ensemble Rejection Sampling” (**Conference on Neural Information Processing Systems 2025**).  
<https://www.arxiv.org/abs/2510.05552>

**Buu Phan\***, Ashish Khisti\*, and Christos Louizos. “Importance matching lemma for lossy compression with side information.” (**International Conference on Artificial Intelligence and Statistics 2024**).  
<https://arxiv.org/abs/2401.02609>

**Buu Phan**, Ashish J Khisti “On List Decoding with Importance Sampling”. (**International Symposium on Information Theory 2025**).  
<https://ieeexplore.ieee.org/abstract/document/11195600>

Salehkalaibar Sadaf\*, **Buu Phan\***, Jun Chen, Wei Yu, and Ashish Khisti: *On the Choice of Perception Loss Function for Learned Video Compression*. (**Conference on Neural Information Processing Systems 2023**).  
**Spotlight in Neural Compression Workshop, International Conference on Machine Learning 2023**.  
<https://arxiv.org/abs/2305.19301>

Sadaf Salehkalaibar, **Buu Phan**, João Atz Dick, Ashish J Khisti, Jun Chen, Wei Yu “Perception Loss Function Adaptive to Rate for Learned Video Compression”. In Machine Learning Compression Workshop, **NeurIPS Workshop 2024**.

Sadaf Salehkalaibar, **Buu Phan**, Ashish Khisti, and Wei Yu. “Rate-Distortion-Perception

Tradeoff Based on the Conditional Perception Measure.” In 2023 **Biennial Symposium on Communications (BSC)**, IEEE, 2023.  
<https://ieeexplore.ieee.org/document/10201822>

**Buu Phan**, Fahim Mannan, and Felix Heide: *Adversarial imaging pipelines*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, (**CVPR 2021**). <https://arxiv.org/abs/2102.03728>

**Buu Phan**, Samin Khan, Rick Salay, and Krzysztof Czarnecki: *Bayesian uncertainty quantification with synthetic data*. In Computer Safety, Reliability, and Security: (**WAISE-SAFECOMP**) 2019 Workshops (**Best paper award**).  
[https://link.springer.com/chapter/10.1007/978-3-030-26250-1\\_31](https://link.springer.com/chapter/10.1007/978-3-030-26250-1_31).

Nicolas Scheiner, Florian Kraus, Fangyin Wei, **Buu Phan**, Fahim Mannan, et al. “Seeing Around Street Corners: Non-Line-of-Sight Detection and Tracking In-the-Wild Using Doppler Radar.” In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. (**CVPR**) 2020.  
<https://arxiv.org/abs/1912.06613>

## OTHER PUBLICATIONS

Samin Khan, **Buu Phan**, Rick Salay, and Krzysztof Czarnecki. “ProcSy: Procedural Synthetic Dataset Generation Towards Influence Factor Studies Of Semantic Segmentation Networks.” **CVPR Workshops** 2019.

Sachin Vernekar, Ashish Gaurav, Taylor Denouden, **Buu Phan**, Vahdat Abdelzad, Rick Salay, and Krzysztof Czarnecki. “Analysis of confident-classifiers for out-of-distribution detection.” **ICLR SafeML Workshop** 2019. <https://arxiv.org/abs/1904.12220>

**Buu Phan**, Rick Salay, Krzysztof Czarnecki, Vahdat Abdelzad, Taylor Denouden, Sachin Vernekar, “Calibrating Uncertainties in Object Localization Task”, **NeurIPS Bayesian Deep Learning Workshop**, 2018. <https://arxiv.org/abs/1811.11210>

Ian Colwell, **Buu Phan**, Shahwar Saleem, Rick Salay, and Krzysztof Czarnecki. “An automated vehicle safety concept based on runtime restriction of the operational design domain.” In **2018 IEEE Intelligent Vehicles Symposium (IV)**.

Denouden Taylor, Rick Salay, Krzysztof Czarnecki, Vahdat Abdelzad, **Buu Phan**, and Sachin Vernekar. “Improving reconstruction autoencoder out-of-distribution detection with mahalanobis distance”, preprint 2018. <https://arxiv.org/abs/1812.02765>

## AWARDS

*Shahid U. H. Qureshi Memorial Scholarship* 2026  
 Award for top students in ECE Department.

*Ontario Graduate Scholarship* 2023  
 Award for top students in Ontario, Canada.

*Best paper award* 2019  
 Received at the Workshop on Artificial Intelligence Safety Engineering for the paper “Bayesian uncertainty quantification with synthetic data.”

*Toyota Canada Graduate Scholarship* 2018  
 Scholarship for graduate students working in AI Safety.

*Faculty of Engineering Awards, University of Waterloo* 2018, 2019

Scholarship for top-performing graduate students.

*International Master's Student Awards, University of Waterloo* 2017-2019  
Scholarship for international graduate students.

*Undergraduate Valedictorian, Vietnam National University (IU Campus)* 2016

**ACADEMIC  
SERVICES**

- Journal Reviewer - TMLR, IEEE.
- Conference Reviewer - AISTATS/CVPR/ICLR/NeurIPS.

**SOFTWARE  
SKILLS**

Programming: PyTorch, HuggingFace, Tensorflow, Matlab, PyThon, C/C++.  
System: Unix, MacOS .  
Tools: Git, Slurm, Docker, WandB.